Length	SAE Rank = 1	SAE Rank = 2	SAE Rank = 3	SAE Rank = 4	SAE Rank = 5
1 (w/o inst.)	1828	717	6919	15917	7056
1 (w/ inst.)	1828	7056	9214	2967	14983
2 (w/o inst.)	1828	717	6919	15917	7056
2 (w/ inst.)	1828	7056	9214	2967	15917
3 (w/o inst.)	1828	717	6919	15917	7056
3 (w/ inst.)	1828	7056	9214	15917	2967

Table 18: Top 1-5 activated SAE features for length constraint

Length	SAE Rank = 6	SAE Rank = 7	SAE Rank = 8	SAE Rank = 9	SAE Rank = 10
1 (w/o inst.)	5549	208	12122	9214	13295
1 (w/ inst.)	15917	5679	2992	3862	1377
2 (w/o inst.)	5549	208	12122	9214	13295
2 (w/ inst.)	1377	717	6919	4371	2992
3 (w/o inst.)	5549	208	12122	9214	13295
3 (w/ inst.)	6919	1377	717	4371	2992

Table 19: Top 6-10 activated SAE features for length constraint

settings, appears to contribute general linguistic structure, while features like SAE_{9214} and SAE_{2967} likely encode finer-grained length or formatting control.

F SAE Feature Similarity

Since the dimension of SAE steering vectors and MeanActDiff steering vectors are the same, we here define the similarity metrics, with either 1) the internal-similarity between two SAE vectors SAE_i and SAE_j :

$$\text{sim_I}(\text{SAE}_i, \text{SAE}_j) = \frac{\langle \text{SAE}_i, \text{SAE}_j \rangle}{\|\text{SAE}_i\|_2 \|\text{SAE}_i\|_2} \quad (7)$$

or 2) the external-similarity between a pair of SAE_i and MeanActDiff_i for each reasoning case:

$$\text{sim_E} = \frac{\langle \text{SAE}_i, \text{MeanActDiff}_i \rangle}{\|\text{SAE}_i\|_2 \|\text{MeanActDiff}_i\|_2} \quad (8)$$

For clarity, we present two similarity matrices showing the top-5 most activated SAE features for the Gemma-2-2B and Gemma-2-9B models in the math reasoning cases respectively. Furthurmore, we provide a table that reports the similarity metric sim_E for Gemma-2-2B and Gemma-2-9B models. This metric quantifies the alignment between the vectors that trigger Chain-of-Thought (CoT) reasoning behavior and those derived from MeanActDiff steering, helping to explain the observed similarities in their effects.

Overall the similarity between a pair of SAE features is low. This makes sense since SAE training tried to disentangle feature representation. Feature SAE_{2899} and feature SAE_{6782} are positively related other four features. SAE_{1858} and SAE_{2899} share high similarity as observed, and in the Neuronpedia we found that they share the common top activated word "The". The similarity between SAE steering vectors and MeanActDiff steering vectors is somehow correlated, which is shown in the Table 22.

F.1 Instruction Following

We visualize the pairwise cosine similarity of selected SAE steering vectors (Figure 19) and MeanActDiff steering vectors (Figure 20), re-using the similarity metrics from section 5.2.3.

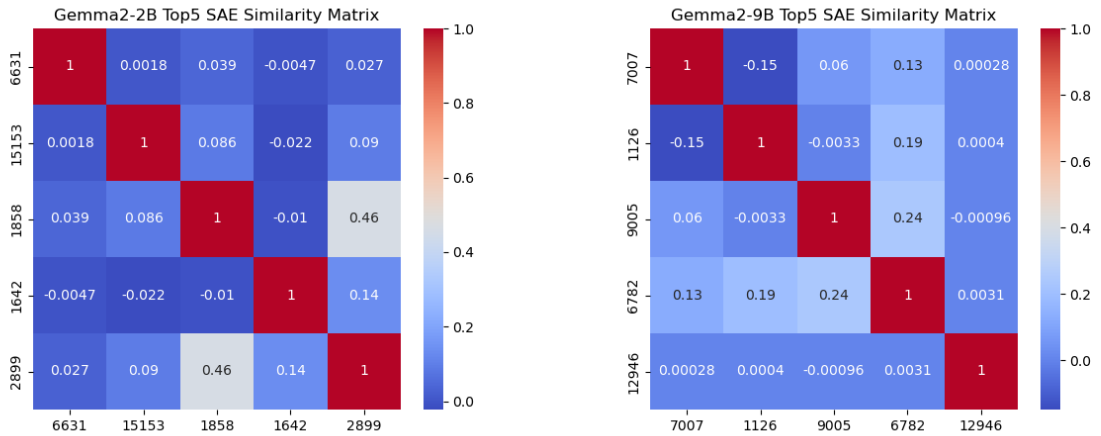
In Figure 19, we observe that the cosine similarity between most SAE features is small and close to zero, suggesting that these features are relatively independent. However, a few pairs such as SAE_{14869} and SAE_{3579}, exhibit moderate similarity (approximately 0.47), indicating some degree of entanglement or functional overlap. SAE_{14869} corresponds

Model Mode	With Instruction	Sentences (Length = 1)	Sentences (Length = 2)	Sentences (Length = 3)	Sentences (Length = 4)
Inference	×	9.30	6.98	9.30	2.33
MeanActDiff	✓	74.42	67.44	60.47	51.16
SAE_{7056}	✓	69.77	58.14	58.14	55.81
SAE_{9214}	✓	69.77	62.79	58.14	58.14
SAE_{2967}	✓	72.09	62.79	60.47	53.49
SAE_{1377}	✓	74.42	62.79	65.12	51.16
Inference	✓	72.09	60.47	58.14	51.16

Table 20: Gemma-2-9B-it: Performance on Instruction Following Dataset IFEval (From At Least to Exactly)

Model Mode	With Instruction	Sentences (Length = 1)	Sentences (Length = 2)	Sentences (Length = 3)	Sentences (Length = 4)
Inference	×	0	4.55	4.55	2.27
MeanActDiff	✓	77.28	70.46	72.73	65.91
SAE_{7056}	✓	65.91	65.91	70.46	63.64
SAE_{9214}	✓	72.73	68.19	63.64	65.91
SAE_{2967}	✓	77.28	72.73	72.73	75.00
SAE_{1377}	✓	70.46	63.64	63.64	70.45
Inference	✓	72.73	72.73	65.91	65.91

Table 21: Gemma-2-9B-it: Performance on Instruction Following Dataset IFEval (From Less Than to Exactly)



(a) Gemma-2-2B Top-5 SAE Feature Similariry Matrix (b) Gemma-2-9B Top-5 SAE Feature Similariry Matrix

Figure 18: Cosine similarity matrices of Top-5 activated SAE features from Gemma-2-2B and Gemma-2-9B models.

Model	Sim_E (max = 1)
Gemma-2-2B	0.2227
Gemma-2-9B	0.3465

Table 22: Cosine similarity between the SAE steering vector that represents CoT reasoning and corresponding MeanActAdd steering vector

to Thai in the SAE latent space, while SAE_{3579} represents Vietnamese. The observed similarity may reflect shared characteristics such as geographical proximity, tonal language structure, and linguistic patterns common to Southeast Asian languages, including subject–verb–object syntax and the use of classifiers.

The MeanActDiff vectors show a much clearer semantic structure. Specifically, steering vectors corresponding to `bn` (Bengali) and `hi` (Hindi) exhibit high cosine similarity (>0.6), which aligns with the linguistic proximity of these languages. This phenomenon suggests that the SAE has learned to group language-related activations in a semantically meaningful way.

We also observe that the vectors for `capital` and `lowercase` have relatively high similarity (>0.5), indicating that these features may encode closely related formatting or syntactic styles. This structured similarity does not emerge as clearly in the raw SAE feature space, further confirming that **MeanActDiff steering surfaces latent semantic**

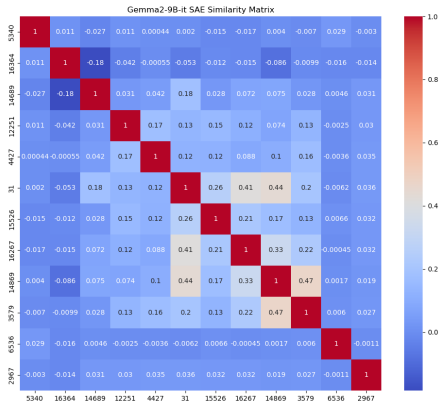


Figure 19: SAE feature similarity matrix

relationships more explicitly, which is the potential reason why MeanActDiff steering is better than SAE steering in lowercase generation.

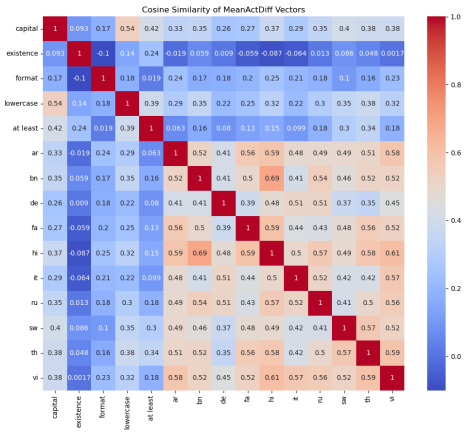


Figure 20: MeanActDiff similarity matrix

Table 23 reports the cosine similarity (`Sim_E`) between the SAE steering vector and its corresponding MeanActAdd steering vector across various instruction cases in the IFEval dataset. A higher similarity indicates that the two steering strategies are capturing consistent directional influence in activation space.

We observe that **format-related tasks**, such as *Uppercase*, *Lowercase*, and *JSON*, all exhibit relatively high similarity scores (ranging from 0.44 to 0.50), suggesting that these syntactic modifications activate similar directions in the model, regardless of the steering method. In contrast, *Length Constraint* matching shows the lowest similarity (0.0985), possibly because it involves stricter token-level constraints that are harder to align in latent space, or because the SAE may not localize the exact-matching behavior as strongly.

Case	Sim_E (max = 1)
Uppercase	0.4938
Lowercase	0.5031
JSON	0.4460
Length Constraint	0.0985
Language-German	0.4649
Language-Italian	0.5209
Language-Hindi	0.3030
Language-Russian	0.3618
Language-Persian	0.4531
Language-Arabic	0.4794
Language-Thai	0.3470
Language-Vietnam	0.3855

Table 23: Cosine similarity between the SAE steering vector and corresponding MeanActAdd steering vector for each case in instruction-followed IFEval dataset

Among language-related instructions, most language pairs exhibit relatively high cosine similarity (typically above 0.45), indicating that the corresponding MeanActDiff vectors activate overlapping regions in the model's representation space. For example, vectors corresponding to *Italian*, *Arabic*, and *Persian* show strong alignment, but even languages such as *Hindi*, *Thai*, and *Vietnamese* maintain moderately high similarity with others. This suggests a shared latent structure across language steering signals. The slight variability may be attributed to differences in tokenization coverage, script diversity, or activation sparsity associated with each language.

G Prompt Template

This section outlines the prompt template used in instruction-following tasks. These templates enforce strict formatting and response structures to encourage model consistency.

For math reasoning prompts, we include 8-shot chain-of-thought examples inside a shaded gray box preceding the query to improve multi-step reasoning: